

Two Simulations for Mystery Tokens

Unit 3 · Topic 3.14 · TODAY'S PROBLEM

Suppose a company claims 10,000 mystery tokens are 50% red, 30% teal, and 20% gold. An SRS of 80 has the shown counts.

Leo: “Mix and redraw these same 80 tokens without replacement; recompute χ^2 for 1,000 trials.”

Amara: “For each trial, generate 80 independent colors with probabilities 0.50, 0.30, and 0.20, then recompute χ^2 .” In Amara’s 1,000 trials, 186 statistics are at least the observed value.

Token sample

Color	Claim	Observed
Red	0.50	32
Teal	0.30	30
Gold	0.20	18

FIRST TAKE (5 min, on your sheet)

Which plan can produce a different number of red tokens on the next trial? Explain in one sentence.

- DOK 1** (a) Find the three expected color counts from the company’s percentages.
- DOK 2** (b) The observed statistic is $\chi^2 = 3.35$. Check the red contribution $(32 - 40)^2/40$, then compute the simulation proportion $186/1,000$.
- DOK 3** (c) In 3–4 sentences, choose a simulation and justify what varies from trial to trial. Use the simulated proportion to decide whether the result is unusual at a 5 percent cutoff, and explain why this does not prove the company’s claim.

Optional review: [follow-along](#) (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.

